

Course Code : 2501CS303**Date** : 14-10-2025**Course Name** : Front-End Web Technology**Duration** : 150 Minutes**Total Marks** : 70**Instructions:**

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 (A) Differentiate between internal and external JavaScript with suitable code snippets. **4**

(B) What are JavaScript pop-up boxes? Explain them with an example. **3**

OR

List and explain any three commonly used JavaScript inbuilt functions with their usage.

(C) Explain event handling in JavaScript with examples. Demonstrate how a callback function can be passed as an argument to another function. **7**

OR

Write a JavaScript program that takes an email and password as input from the user and validates them using regular expressions. The program should ensure that the email address is valid and the password is at least 8 characters long, containing at least one uppercase letter and one special symbol.

Q.2 (A) Explain jQuery selectors with suitable examples. **4**

(B) What is an arrow function? Write a simple example. **3**

OR

Explain ES6 destructuring with suitable examples.

(C) With suitable examples, explain the use of spread and rest operators in ES6. **7**

OR

List any six ES6 array methods and explain the map and reduce methods with a suitable example.

Q.3 (A) What is JSX in React? Explain its usage in detail and also provide a small demo program to illustrate it. **4**

(B) Explain the difference between functional components and class components in React. **3**

OR

Explain the difference between state and props with examples.

- (C) Explain how React handles conditional rendering using both if-else statements and the ternary operator. Write a sample component that demonstrates both methods. **7**

OR

Explain how event handling works in React. Write a React program that demonstrates handling at least two different types of events on components.

- Q.4 (A)** Explain React Routing with a suitable example. **4**
- (B)** What is the purpose of the useEffect hook? Explain with an example. **3**

OR

What is the purpose of the useParams hook? Explain with an example.

- (C) Demonstrate how to consume an API in React to perform GET all, GET by ID, and DELETE operations. **7**

OR

Demonstrate how to consume an API in React to perform GET all and UPDATE operations.

- Q.5 (A)** Differentiate API authentication and API authorization with examples. **4**
- (B)** Define TLS encryption and explain its role in API security. **3**

OR

Explain JWT (JSON Web Token) and its components.

- (C) Explain OAuth 2.0 in detail. **7**

OR

Discuss the best practices for consuming an API in detail.
